



EAD Variance Analyzer with LangChain

Upload two datasets (old vs new) to analyze EAD changes and get an AI explanation.

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inputs_data1_v4.xlsx 22.9KB



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summarize in table the top 3 EAD changes



Top 3 EAD Changes Summary

Based on the EAD comparison data, here are the top 3 netting sets with the largest EAD changes:

Rank	Netting ID	Old EAD	New EAD	EAD Change (Δ)	% Change	Key Driver
1	MNA001	621,365	1,366,947	+745,582	+120.0%	Replacement Cost increased from 0 to 238,958 due to VM reduction from 1,300,000 to 300,000; Multiplier increased from 0.60 to 1.00
2	MNA002	2,901,185	2,923,999	+22,814	+0.79%	PFE addon increased by 16,295 due to Trade_005 maturity extension (end_date: 199→499 days, Maturity: 199→499 days)
3	MNA003	2,304,026	2,325,316	+21,291	+0.92%	Replacement Cost increased by 15,208 due to market value changes in Trade_001 (+24,390) and Trade_002 (-9,182)

Key Observations:

- **MNA001** experienced the most dramatic change, with EAD nearly doubling due to a significant reduction in variation margin (VM) collateral, which increased the replacement cost and eliminated the multiplier

dampening effect

- **MNA002** and **MNA003** had modest increases driven by trade maturity extensions and mark-to-market movements respectively
- All other netting sets (MNA004, MNA005, MNA006, MNA007) showed no EAD changes between old and new datasets



Verify supervisory delta for the netting sets



Supervisory Delta Verification Analysis

Overview

I have reviewed the supervisory delta calculations across all netting sets and identified a significant discrepancy for one trade in netting set MNA006.

Discrepancy Identified

Netting Set MNA006 - Trade_001:

- **System Delta Value:** -3.0
- **Calculated Delta Value:** -0.64617
- **Absolute Difference:** 2.35383
- **Status:** ⚠️ **VALIDATION FAILED**

Root Cause Analysis

The discrepancy stems from the delta calculation methodology for this specific trade. Based on the calculated delta value of -0.64617, this indicates:

1. **Transaction Type:** The calculated delta suggests this is a **short position** in a derivative instrument (negative delta indicates short exposure)
2. **Delta Calculation Logic:** The calculated value of -0.64617 falls within the typical range for:
 - Option positions (where delta ranges between -1 and +1)
 - The magnitude suggests this could be an at-the-money or slightly out-of-the-money option position
3. **System Override Issue:** The system delta of -3.0 appears to be incorrect as:
 - Standard supervisory delta values under SA-CCR should not exceed ± 1.0 for non-CDO trades
 - For CDO tranches, the maximum delta magnitude is capped at $\pm 15 / ((1 + 14 \times A) \times (1 + 14 \times D))$ which typically produces values well below 3.0
 - A delta of -3.0 suggests either a data entry error or an incorrect calculation methodology in the source system

Impact Assessment

EAD Impact for MNA006:

- The supervisory delta directly multiplies the adjusted notional and supervisory factor in the trade add-on calculation

- An inflated delta magnitude of -3.0 versus -0.64617 would result in:
 - **Trade add-on overstatement** by a factor of approximately 4.64x ($3.0/0.64617$)
 - This would cascade through to the hedging set amount
 - Ultimately inflating the PFE add-on component of EAD

Current EAD Status for MNA006:

- Calculated EAD: 18,534.75
 - System EAD: 578,290.50
 - The significant difference between calculated and system EAD may be partially attributable to this delta discrepancy
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Recommendations

1. **Immediate Action:** Investigate the source data for Trade_001 in netting set MNA006 to determine why the system delta is set to -3.0
2. **Data Validation:** Review the trade characteristics:
 - Confirm if this is an option trade (check is_option flag)
 - Verify underlying price, strike price, and option maturity inputs
 - Check if CDO tranche flags are correctly set
3. **System Correction:** Update the system delta to align with the calculated value of -0.64617 or investigate if there are special circumstances warranting the -3.0 value
4. **Broader Review:** Conduct a systematic review of all supervisory delta values across the portfolio to identify any other anomalies

All Other Trades: VALIDATED

All remaining trades across netting sets MNA001, MNA002, MNA003, MNA004, MNA005, and MNA007 show supervisory delta values that align with calculated values, indicating proper delta calculation methodology for the rest of the portfolio.

Enter a question...

